

Last Time:

- $\mathbb{R}$ = set of real numbers

- $\mathbb{R}^2 \stackrel{\text{LHS}}{:=} \{ \underbrace{(a, b)}_{\text{a pair = 2-tuple}} : \underbrace{a, b \in \mathbb{R}}_{\text{means } a \& b \text{ are elements of } \mathbb{R}} \}$

(Left-hand side is defined by Right-hand side)

- Cartesian plane is a way to visualize $\mathbb{R}^2$

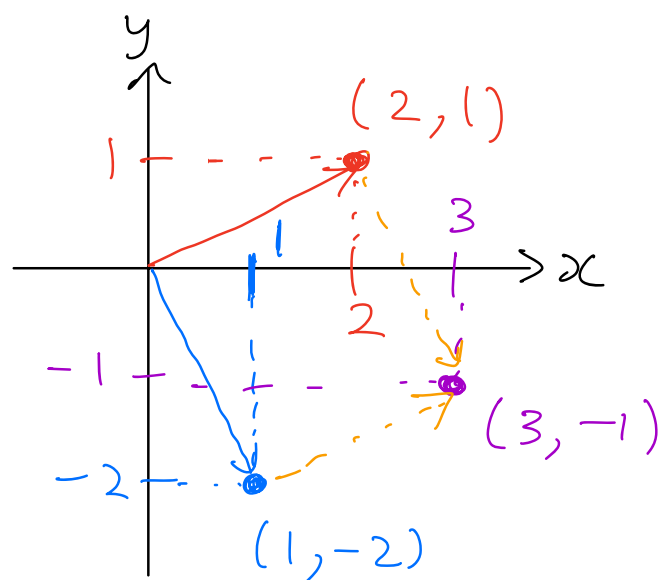
Today: Continue §1.1 $\mathbb{R}^n$ as a Vector Space

There are two operations on $\mathbb{R}^2$:

(1) Coordinate-wise Addition: If $(a_1, a_2), (b_1, b_2) \in \mathbb{R}^2$,
then $(a_1, a_2) + (b_1, b_2) := (a_1 + b_1, a_2 + b_2) \in \mathbb{R}^2$

For example,

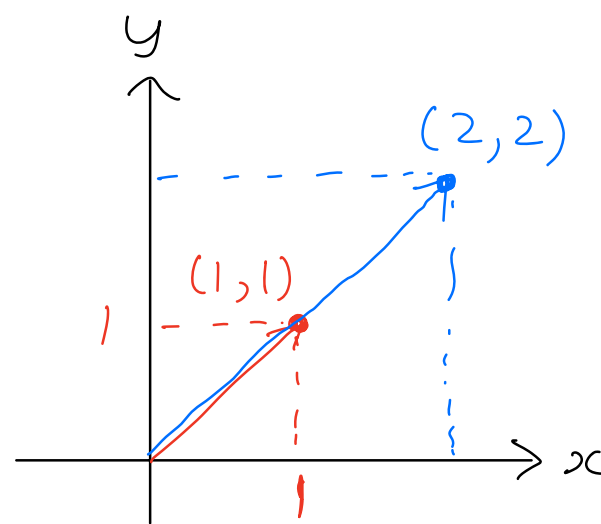
$$\begin{aligned} (2, 1) + (1, -2) &= (2+1, 1-2) \\ &= (3, -1) \end{aligned}$$



(2) Scalar Multiplication: If $c \in \mathbb{R}$ and $(x, y) \in \mathbb{R}^2$,
then $c \cdot (x, y) := (cx, cy)$.

For example,

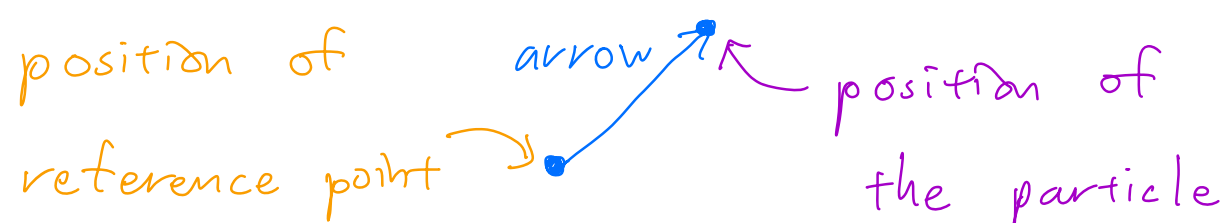
$$2 \cdot (1, 1) = (2 \cdot 1, 2 \cdot 1) = (2, 2)$$



Note: Subtraction on $\mathbb{R}^2$ is

$$\begin{aligned} (a, b) - (c, d) &:= (a, b) + (-1) \cdot (c, d) = (a, b) + (-c, -d) \\ &= (a - c, b - d) \end{aligned}$$

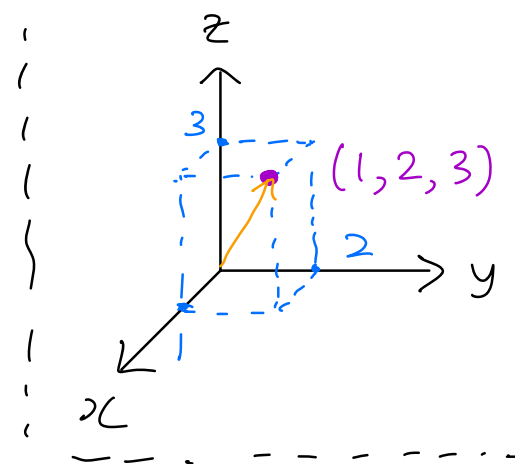
Rmk: In Physics, displacement (relative position of a particle from fixed reference point) on a plane can be modelled as a point of $\mathbb{R}^2$



Similarly, $\mathbb{R}^3$ has many properties similar to $\mathbb{R}^2$:

Def 1.2: $\mathbb{R}^3 := \{(a, b, c) : a, b, c \in \mathbb{R}\}$

3-tuple of real numbers



$\mathbb{R}^3$ has both $+$ & $\cdot$:

$$(a_1, a_2, a_3) + (b_1, b_2, b_3) := (a_1 + b_1, a_2 + b_2, a_3 + b_3)$$

$$c \cdot (a_1, a_2, a_3) := (ca_1, ca_2, ca_3)$$

whenever $a_1, a_2, a_3, b_1, b_2, b_3, c \in \mathbb{R}$.

$$\left(\begin{aligned} \text{and } (a_1, a_2, a_3) - (b_1, b_2, b_3) \\ &:= (a_1, a_2, a_3) + (-1) \cdot (b_1, b_2, b_3) \\ &= (a_1 - b_1, a_2 - b_2, a_3 - b_3) \end{aligned} \right)$$

Mathematician's

Guiding Principle: when people say a "vector", they mean "things" that can be added & scalar multiplied. Similarly, "vector space" means a set of "vectors" (with additional properties). In this sense, both $\mathbb{R}^2$ & $\mathbb{R}^3$ are vector spaces. (We will precisely define these words in Section 1.2)

$\mathbb{R}^2$ represents 2-dimensional plane and $\mathbb{R}^3$ represents 3-dimensional space. So to represent higher-dimensional space (e.g. spacetime), need to generalize $\mathbb{R}^2$ & $\mathbb{R}^3$ into $\mathbb{R}^n$:

Def 1.3: Fix $n \in \mathbb{Z}_{>0} := \{1, 2, 3, \dots\}$. Then

$$\mathbb{R}^n := \{(\underbrace{x_1, \dots, x_n}_{\text{n-tuple of real } \mathbb{R} \text{ \#s}}) : x_i \in \mathbb{R} \text{ for every } i=1, \dots, n\}.$$

n-tuple of real $\mathbb{R}$ #s

Given $(x_1, \dots, x_n) \in \mathbb{R}^n$, its $\underbrace{k^{\text{th}}}_{\text{coord}} - \text{coordinate}$ is x_k .

Rmk: Some books def $\mathbb{R}^0 := \{0\}$.

Def 1.4: Addition and scalar-multiplication on

$\mathbb{R}^n$ is:

$$(1) \forall (x_1, \dots, x_n), (y_1, \dots, y_n) \in \mathbb{R}^n,$$

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) := (x_1 + y_1, \dots, x_n + y_n) \in \mathbb{R}^n$$

$$(2) \forall (x_1, \dots, x_n) \in \mathbb{R}^n \text{ \& } \forall c \in \mathbb{R},$$

$$c \cdot (x_1, \dots, x_n) = c(x_1, \dots, x_n) := (cx_1, \dots, cx_n)$$

To extract properties of $\mathbb{R}^n$, we need to re-interpret $+$ & $\cdot$ via sets and functions (also called maps):

Def: (Cartesian) product of two sets X, Y is

$$X \times Y := \{ (x, y) : x \in X, y \in Y \}.$$

$$\boxed{\begin{array}{l} f: X \rightarrow Y \text{ function} \\ x \mapsto f(x) \text{ } x \text{ maps to } f(x) \end{array}}$$

Example: $\mathbb{R}^2 = \mathbb{R} \times \mathbb{R}$, $\mathbb{R}^3 \xrightleftharpoons[\text{1-1 con}]{\text{1-1 onto}} \mathbb{R}^2 \times \mathbb{R}$
 $(a, b, c) \mapsto ((a, b), c)$

Note 1.5: Def 1.4 & Prev Def

$\Rightarrow + \text{ \& } \cdot$ are ^{funcs} functions as well:

$$+ : \mathbb{R}^n \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

$$((x_1, \dots, x_n), (y_1, \dots, y_n)) \mapsto (x_1 + y_1, \dots, x_n + y_n)$$

$$\cdot : \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

$$(c, (x_1, \dots, x_n)) \mapsto (cx_1, \dots, cx_n)$$

$\Downarrow$

Proposition

Prop 1.6: $\mathbb{R}^n$ ^{w/} with $+ : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\cdot : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ ^{vector addition} satisfying ^{set of scalars} ^{scalar-multi}

$\cdot : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ satisfying:

$$(VS 1) \forall v, w \in \mathbb{R}^n, v + w = w + v$$

$$(VS 2) \forall u, v, w \in \mathbb{R}^n, (u + v) + w = u + (v + w)$$

(VS 3) There $\exists$ exists $0_{\mathbb{R}^n} \in \mathbb{R}^n$ (or simply called 0)

such that ^{s.t.} $\forall v \in \mathbb{R}^n, v + 0_{\mathbb{R}^n} = v = 0_{\mathbb{R}^n} + v$

$$(VS4) \forall v \in \mathbb{R}^n, \exists w \in \mathbb{R}^n \text{ s.t. } v + w = 0_{\mathbb{R}^n}$$

$$(VS5) \forall v \in \mathbb{R}^n, 1 \cdot v = v$$

$$(VS6) \forall a, b \in \mathbb{R} \text{ and } \forall v \in \mathbb{R}^n, (ab) \cdot v = a \cdot (b \cdot v) \quad \text{with } (ab)v \text{ and } a(bv) \text{ also valid}$$

$$(VS7) \forall c \in \mathbb{R} \text{ and } \forall v, w \in \mathbb{R}^n, c(v + w) = cv + cw$$

$$(VS8) \forall a, b \in \mathbb{R} \text{ and } \forall v \in \mathbb{R}^n, (a + b)v = av + bv$$

Remark

Rmk 1.7:

(a) (VS1) – (VS8) tell us that $+$ & $\cdot$ behave “properly” (matches w/ our intuition)

(b) $0_{\mathbb{R}^n} \in \mathbb{R}^n$ from (VS3) is [!]unique.

$0_{\mathbb{R}^n} = (0, \dots, 0) \stackrel{\text{abuse of notation}}{=} 0$ is called a zero vector

(c) From (VS4), the $w \in \mathbb{R}^n$ is just $(-1)v$.

We often write $-v$ as $(-1)v$. So that w is unique.

Idea
Pf of Prop 1.6: ^{b/c} Because $\mathbb{R} = \mathbb{R}^1$ satisfies

(VS1) – (VS8) (Korean high school math) and

$+: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ & $\cdot: \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}^n$ is def

coordinate-wise, $n=1$ case “implies” $n>1$ case.

this needs full-justification



